MSc subjects offered by the Department of Electrical and Electronic Engineering
(MSc in MQ)

MSc in Microelectronics and Quantum Systems Engineering					2024/25*			2025/26*		
Subjects		Without stream option	Elective Streams							
			Design Engineering	Quantum Systems Engineering	Semester 1	Semester 2	Summer Semester	Semester 1	Semester 2	Summer Semester
EEE501	Microelectronic Devices and Fabrication#				D			D		
EEE502	The Physics of Semiconductor#				E			E		
EEE503	Microphotonics and Quantum Photonic Systems#				E				E	
EEE504	Wafer Fabrication and Laboratory#					D			E	
EEE505	Design and Simulation of Photonic Circuits#					D		D		
EEE506	Introduction to Wide-bandgap Semiconductors#					D			D	
EEE508	VLSI Technology and Design	EEE508	EEE508			E			E	
EEE510	Introduction to Quantum Physics	EEE510		EEE510				E		
EEE511	Fundamental Quantum Information	EEE511		EEE511					E	
EEE512	Quantum Optics	EEE512		EEE512					D / E	
EEE513	Introduction to Quantum Computing	EEE513		EEE513		E				
EEE514	Quantum Communication	EEE514		EEE514	E			E		
EEE515	Lasers	EEE515		EEE515		E			E	
EIE515	Advanced Optical Communication Systems	EIE515		EIE515		D			E	
EEE516	Research Project	EEE516	EEE516	EEE516	D	D		D	D	
EIE577	Optoelectronic Devices	EIE577			D			D		
EIE580	RF and Microwave Integrated Circuits for Communication System Applications	EIE580	EIE580		E			D		
LGT5426	Managing Innovation		LGT5426		E	E		E		
AP5007	MEMS (Microelectromechanical Systems) and Sensors		AP5007							D
EEE5T03	Engineering Ethics and Academic Integrity				D / E (For **Full-time** students)	D / E (For **Part-time** students)		D / E (For **Full-time** students)	D / E (For **Part-time** students)	

D Daytime Class
E Evening Class

Core subjects for MSc in MQ

*The Department reserves the right of NOT offering any one of the above subjects. The offer of subjects is subject to review and changes if deemed appropriate by the Department.

(last updated in Jan 2026)